

AI-Based Virtual Assistant for Seamless System Control and Web Interaction

MAJOR PROJECT

Submitted by

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Under the supervision of

Dr. Saoud Sarwar



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DECLARATION

I, Yagsha Rafat a student of **Bachelor of Technology in Computer Science and Engineering (B.Tech CSE)**, (Enrollment No: **2021-310-237**) hereby declare that the project entitled “**AI-Based Virtual Assistant for Seamless System Control and Web Interaction**” which is being submitted by me to the Department of Computer Science & Engineering, School of Engineering Sciences & Technology, Jamia Hamdard, New Delhi in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (B.Tech CSE)**, is my original work and has not been submitted anywhere else for the award of any Degree, Diploma, Associateship, Fellowship or other similar title or recognition.

I declare that this project is the result of my independent effort and has been completed under the guidance of **Dr. Saoud Sarwar**.

Yagsha Rafat

Date:

Place: Jamia Hamdard, New Delhi

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of my project, “**AI-Based Virtual Assistant for Seamless System Control and Web Interaction.**”

I am sincerely thankful to **Dr. Saoud Sarwar**, my project supervisor, for his continuous support, valuable guidance, and encouragement throughout the course of this project.

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I am grateful to my friends and classmates for their cooperation and feedback during the development phase.

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Yagsha Rafat

ABSTRACT

The increasing need for intelligent personal assistants has led to the development of AI-based virtual assistants that can interact with users and perform various tasks. This project, titled **“AI-Based Virtual Assistant for Seamless System Control and Web Interaction”** aims to develop a virtual assistant that can manage both system-level tasks and web-based functions through voice and text commands. Built using Python, OpenCV, and Mediapipe, this virtual assistant offers features such as voice/text command recognition, controlling system functions (e.g., checking the date/time, opening apps, managing files), web searches, and weather updates. Additionally, it includes extra features such as reminders, jokes, notes, and games like Tetris and Snake to improve the user experience.

The assistant uses a strong command-handling system that allows users to easily perform tasks. It also incorporates error handling to manage invalid or negative commands, ensuring smooth operation. With the integration of the OpenWeatherMap API, the assistant can fetch real-time weather data. Other features, such as speech-to-text transcription and camera controls, make the assistant even more versatile.

The goal of this project is to create an intelligent, efficient, and user-friendly virtual assistant that simplifies everyday tasks and helps improve productivity by offering both system-level and web interaction support. The assistant is designed to be flexible, allowing for the addition of more features in the future.

LIST OF ABBREVIATIONS

S. No	Abbreviation	Full Form
1.	AI	Artificial Intelligence
2.	VA	Virtual Assistant
3.	GUI	Graphical User Interface
4.	NLP	Natural Language Processing
5.	STT	Speech to Text
6.	TTS	Text to Speech
7.	OS	Operating System
8.	API	Application Programming Interface
9.	CMD	Command Prompt
10.	CPU	Central Processing Unit
11.	RAM	Random Access Memory
12.	UI	User Interface
13.	UX	User Experience
14.	UML	Unified Modeling Language
15.	ERD	Entity Relationship Diagram
16.	SRS	Software Requirements Specification
17.	IP	Internet Protocol
18.	ML	Machine Learning
19.	IO	Input/Output

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Chapter-1

Introduction

1.1 Overview

The emergence of artificial intelligence has revolutionized the way humans interact with technology. This project, AI-Based Virtual Assistant for Seamless System Control and Web Interaction, focuses on building an intelligent system that can understand user commands (voice/text) and perform a variety of tasks—ranging from system-level operations to internet-based searches. By integrating speech recognition, system controls, and web APIs, this assistant acts as a multipurpose tool to enhance productivity and accessibility, especially for everyday users.

1.2 Background

Virtual assistants like Siri, Google Assistant, and Alexa have set the benchmark for intelligent automation and voice-based control. These systems rely on advanced natural language processing and machine learning algorithms. Inspired by these technologies, this project aims to create a personalized assistant that not only understands user inputs but also performs tasks like launching applications, checking the weather, taking notes, opening games, converting speech to text, setting reminders, and interacting with online platforms. The background study includes an understanding of Python libraries like SpeechRecognition, pyttsx3, OpenCV, and APIs like OpenWeatherMap and joke/news APIs.

1.3 Motivation

In the digital era, users often juggle between multiple applications and websites to complete simple tasks. This not only consumes time but also affects workflow efficiency. The motivation behind this project is to provide users with a unified AI-powered assistant that understands both text and speech, and simplifies common actions through a single command interface. Additionally, such a system can be particularly beneficial for individuals with mobility limitations or those who prefer voice-controlled environments.

1.4 Problem Statement

Despite the widespread use of virtual assistants like Siri, Google Assistant, and Alexa, most systems are either platform-dependent or fail to provide sufficient customization for local system control. Additionally, many existing virtual assistants are often limited to specific tasks or services and lack the ability to execute system-level commands or integrate seamlessly with desktop environments.

There is a gap in creating a flexible, Python-based virtual assistant that can operate across various desktop platforms (Windows, Linux, macOS), understand both voice and text commands, and perform a wide range of tasks, from system-level functions to web-based operations. This project addresses this gap by developing a lightweight, customizable virtual assistant that bridges the interaction between users and their machines, making it easier to manage and control both local systems and web-based tasks effectively.

1.4.1 Platform Dependency

Discuss how many current virtual assistants are confined to specific operating systems or ecosystems (e.g., Apple's Siri, Amazon's Alexa).

1.4.2 Limited Functionality

Point out the restrictions of existing virtual assistants in terms of system control (e.g., inability to interact with desktop applications or control local system settings).

1.4.3 Lack of Customization

Mention how most virtual assistants do not offer the flexibility to tailor them to specific user needs or preferences, especially in desktop environments.

1.4.4 Need for Seamless Integration

Emphasize the need for a virtual assistant that integrates effectively with both system-level commands and online platforms to perform various tasks.

1.5 Objective

The primary objectives of this project are:

- To design and develop an intelligent AI-based virtual assistant using Python programming language.
- To integrate both voice and text input methods for flexible command interaction.
- To enable the assistant to perform system-level tasks such as launching applications, checking the date and time, taking screenshots, opening file explorer, and capturing photos using the webcam.
- To support web-based functionalities, including fetching live weather reports, telling jokes, retrieving news headlines, and conducting Google or YouTube searches.
- To implement utility features like speech-to-text conversion, reminder setting, note-taking, and game launching.
- To allow the assistant to understand and respond to negative commands (e.g., “don’t show time” or “don’t open Chrome”).
- To ensure the assistant provides a seamless, interactive, and user-friendly experience for desktop users.
- To build a lightweight, customizable, and locally executable solution that does not rely heavily on external services.

1.6 Scope of the Project

The scope of this project extends to the development of a fully functional AI-based virtual assistant capable of performing a wide range of system-level and internet-based tasks using both voice and text commands. The assistant is designed to enhance user productivity and ease of access by providing a hands-free and intuitive interface for desktop environments.

Key areas covered in the project scope include:

1.6.1 Voice and Text Input Handling

The assistant will accept commands through both voice and text, allowing flexible user interaction.

1.6.2 System-Level Control

It will manage local operations such as launching installed applications (e.g., Chrome, Notepad), checking the date and time, capturing screenshots, opening file explorer, and using the system's webcam.

1.6.3 Web-Based Features

The assistant will be integrated with external APIs to provide real-time weather updates, news headlines, jokes, and allow Google/YouTube search capabilities.

1.6.4 Utility Modules

It includes productivity features such as speech-to-text document writing, note creation, setting reminders, and opening pre-integrated games like Snake and Tetris.

1.6.5 Natural Command Handling

The system can understand and appropriately handle negative commands (e.g., “don't open browser” or “don't show time”).

1.6.6 Customization and Portability

The assistant is built using Python and open-source libraries, making it lightweight, customizable, and compatible with various Linux-based desktop systems.

Chapter-2

System Requirements and Specifications

2.1 System Requirements

The development and operation of the AI-Based Virtual Assistant for Seamless System Control and Web Interaction require a set of essential system requirements to ensure functionality, performance, and compatibility. These requirements are categorized into hardware, software, functional, and non-functional components.

2.1.1 Hardware Requirements

To successfully execute the assistant and handle multimedia and real-time functionalities, the following hardware is recommended:

- A personal computer or laptop with a minimum Intel Core i5 processor or equivalent.
- At least 8 GB RAM to ensure efficient multitasking and smooth execution of commands.
- A minimum of 256 GB of internal storage (HDD or SSD) for application data, logs, and user-generated files.
- A functional microphone (built-in or external) for capturing voice commands.
- Headphones or speakers for audio responses.
- Webcam (optional) for camera capture feature.
- Stable internet connection for API-based services such as weather updates, jokes, and news.

2.1.2 Software Requirements

The system requires a stable software environment that supports Python-based development and open-source libraries. The recommended software stack includes:

- **Operating System:** Linux (Ubuntu/Debian) or Windows 10 and above.
- **Programming Language:** Python 3.8 or higher.
- **Essential Libraries and Modules**
 - speech_recognition, pyttsx3, pyautogui, requests, datetime, subprocess, pygame, webbrowser, opencv-python, json, tkinter, threading, os, psutil, pywhatkit.
- **API Services Used**
 - OpenWeatherMap API (for live weather).[6]
 - JokeAPI (for real-time jokes).[7]
- **Other Tools**
 - Visual Studio Code or PyCharm (IDE).
 - Chrome or Firefox browser.
 - GitHub (for version control and collaboration).
 - Anaconda (optional, for virtual environments).

2.1.3 Functional Requirements

The assistant is built to perform a variety of system-level and web-based tasks based on user input. The core functional requirements are as follows:

- Accept input via both voice and text modes.
- Convert text responses into speech for vocal interaction.
- Fetch and announce system details like current date, time, and battery status.
- Recognize and respond appropriately to negative commands (e.g., “don’t show time”).
- Open and close commonly used software and websites (e.g., Chrome, Notepad, LinkedIn, GitHub).
- Perform online searches and display or speak the result.
- Retrieve real-time weather data and tell current weather conditions.
- Fetch jokes from an online source and vocalize them.
- Capture screenshots and photos using a system camera.
- Convert user speech to text and save it as a document.
- Launch mini-games such as Snake and Tetris.
- Open image galleries or file explorers.
- Create and store quick notes locally.
- Set reminders and alert the user at specified times.

2.1.4 Non-Functional Requirements

Apart from the direct functionalities, the system must adhere to several quality standards for effective user experience. These include:

- **Usability:** The assistant should be intuitive to operate through a clean and simple interface.
- **Reliability:** It should deliver consistent performance without crashes or errors.
- **Performance:** Commands should be processed and executed with minimal delay.
- **Scalability:** The system should be extensible, allowing future modules (e.g., mobile app, gesture control).
- **Security:** All user data, notes, and reminders should remain confidential and locally stored.
- **Portability:** The application should run across platforms like Linux and Windows with minimal changes.

- **Maintainability:** The system should be modular with clear documentation for future updates.

2.2 System Specifications

The AI Virtual Assistant has been designed with modularity and user-centric control in mind. The specifications of the system are as follows:

- The assistant offers both text and voice command support, ensuring flexible interaction.
- It greets the user and responds audibly using a TTS engine (pyttsx3).
- The system provides real-time feedback on basic system metrics like date, time, and battery.
- The application intelligently parses commands and skips tasks based on negative phrasing.
- It enables launching and control of desktop applications through subprocess calls and system commands.
- Internet-dependent features such as weather, news, and jokes are enabled via API integrations.
- The system supports screenshot capturing and camera activation using pyautogui and cv2.
- Users can dictate content that is transcribed and stored as .txt files automatically.
- It allows for basic gameplay experiences to be launched on request.
- Reminders and notes are handled internally, stored in local files, and triggered using threads and timers.
- The backend is built in Python with modular architecture allowing ease of debugging and feature enhancement.
- The assistant is lightweight and can function offline for most tasks (except online API-based features).
- It is fully customizable for the user's preferences and voice interaction flow.

2.3 Cost and Effort Estimation

The AI Virtual Assistant project is developed as an academic final-year submission, leveraging open-source tools and existing hardware resources. The associated cost and time investments are minimal yet structured as follows:

- **Hardware Cost:** No additional hardware cost incurred as existing personal laptop and peripherals were used.
- **Software Cost:** All tools and libraries used are open-source and free of cost (Python, APIs, libraries).
- **API Usage:** Free tier services of OpenWeatherMap and JokeAPI have been used, eliminating subscription charges.
- **Development Timeframe:**
 - Requirement Gathering and Research: 1 week
 - System Design and Planning: 1 week
 - Development and Integration: 3 weeks
 - Testing and Debugging: 1 week
 - Final Documentation and Review: 1 week
 - **Total Estimated Time:** ~7 weeks (~120–140 hours total effort)
- **Team:** Single developer (Yagsha Rafat)

The project delivers maximum utility at zero monetary cost using a time-efficient and modular approach.

Chapter-3

System Design

This chapter presents the visual and structural design of the AI-Based Virtual Assistant using Use Case, Class, ER, and Activity diagrams to represent system interactions and architecture.[1]

3.1 Use Case Diagram

The Use Case Diagram represents the functional requirements of the virtual assistant system and its interaction with the user. The user interacts with the system through either voice or text commands to perform various actions such as retrieving system information, launching applications, setting reminders, and more.[5]

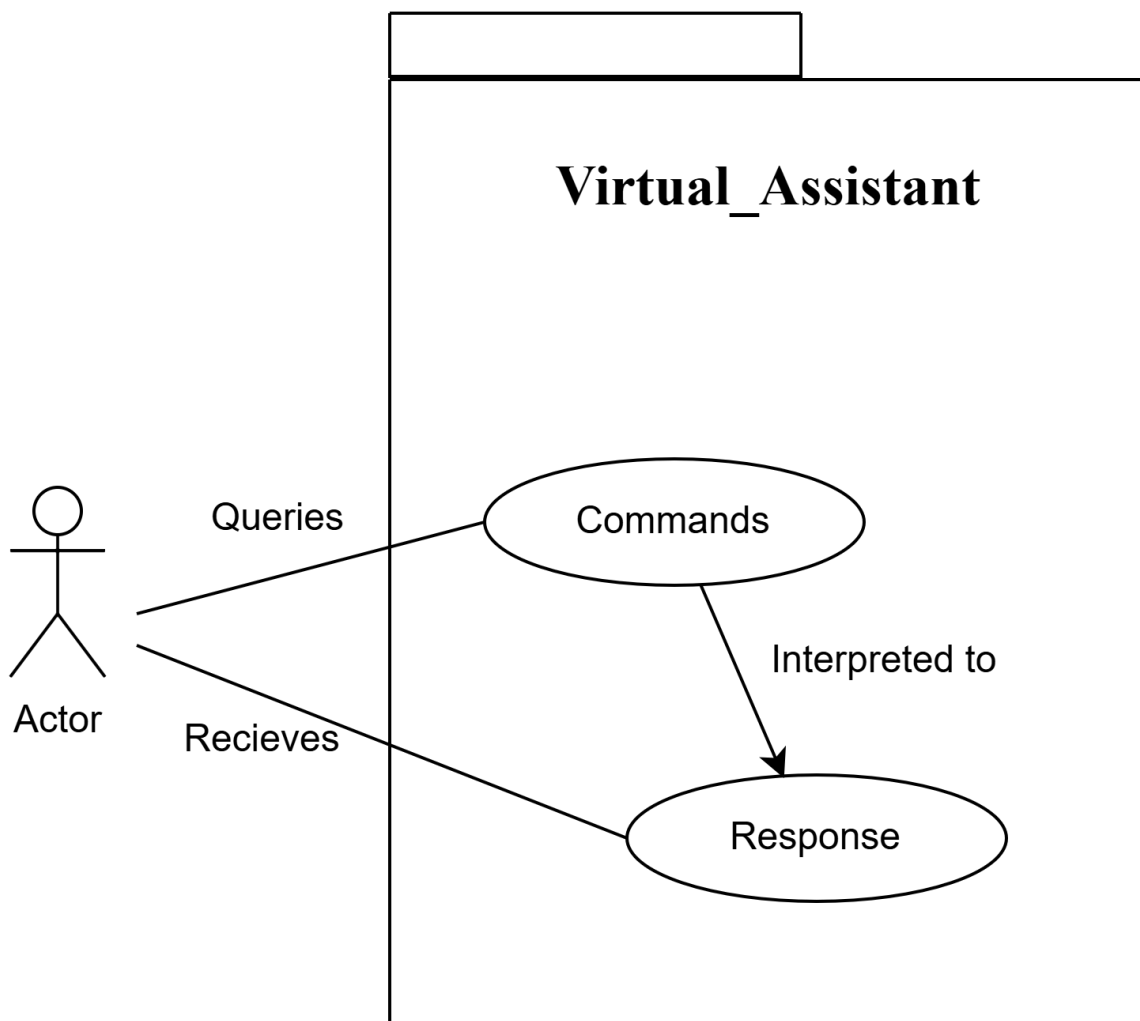


Figure 3.1 Use Case Diagram

3.2 Use Case Specifications (Tabular Format)

Each use case should be specified in a table, detailing how it operates under different scenarios. Below is an example of how the use case for getting the date/time would look.

3.2.1 Use Case: Get Date and Time

Use Case Name	Get Date and Time
Pre-condition	The system must be running.
Basic Path	User issues the command → Assistant fetches and responds with current date/time.
Alternate Path	User types the command instead of speaking.
Post-condition	Date/Time is displayed or spoken.
Exception Path	If microphone access fails → display error message.

3.2.2 Use Case : Launch Application (e.g., Chrome)

Use Case Name	Launch Application (e.g., Chrome)
Pre-condition	Application must be installed.
Basic Path	User says "Open Chrome" → System launches Chrome.
Alternate Path	User types "Launch Chrome".
Post-condition	Application opens successfully.
Exception Path	If application not found → assistant notifies the user.

3.2.3 Use Case : Speech-to-Text Document Writer

Use Case Name	Speech-to-Text Document Writer
Pre-condition	Assistant is running and microphone is active.
Basic Path	User says "Start dictation" → Speaks → Text is saved.
Alternate Path	If user pauses, partial text is saved.
Post-condition	A text file is generated with transcription.
Exception Path	If mic access is denied → assistant shows a warning.

3.2.4 Use Case : Set Reminder

Use Case Name	System is active.
Pre-condition	User says "Set a reminder for 5 PM to call John" → Reminder is saved and scheduled.
Basic Path	User says "Set a reminder for 5 PM to call John" → Reminder is saved and scheduled.
Alternate Path	User provides time and tasks in multiple steps.
Post-condition	Reminders are triggered at the specified time.
Exception Path	Invalid time format → assistant prompts correction.

3.2.5 Use Case : Tell Joke

Use Case Name	Tell Joke
Pre-condition	The assistant must be running.
Basic Path	User says "Tell me a joke" → Assistant fetches and tells a joke from the API.
Alternate Path	User types "Tell me a joke".
Post-condition	Jokes are displayed or spoken.
Exception Path	If the joke API is unavailable → assistant shows an error message.

3.2.6 Use Case : Fetch Weather Information

Use Case Name	Fetch Weather Information
Pre-condition	The system must have internet access.
Basic Path	User asks "What's the weather?" → Assistant fetches weather details from the OpenWeatherMap API.
Alternate Path	User specifies a location, e.g., "What's the weather in New York?"
Post-condition	Weather information is displayed.
Exception Path	If weather data cannot be fetched → Assistant shows an error message.

3.2.7 Use Case : Play Game (e.g., Snake)

Use Case Name	Play Game (e.g., Snake)
Pre-condition	The game should be available in the system.
Basic Path	User says "Play Snake" → Game launches.
Alternate Path	User types "Play Snake".
Post-condition	Game is started successfully.
Exception Path	If the game is not installed → Assistant notifies the user.

3.2.8 Use Case : Open Gallery

Use Case Name	Open Gallery
Pre-condition	Gallery app must be installed.
Basic Path	User says "Open Gallery" → Gallery app opens.
Alternate Path	User types "Open Gallery".
Post-condition	Gallery is opened.
Exception Path	If Gallery app is not found → Assistant notifies the user.

3.2.9 Use Case : Get News

Use Case Name	Get News
Pre-condition	Internet access is required.
Basic Path	User asks "What's the latest news?" → Assistant fetches news from the browser.
Alternate Path	User specifies the topic, e.g., "Show me sports news".
Post-condition	Latest news headlines are displayed.
Exception Path	If news can't be fetched → Assistant shows an error message.

3.2.10 Use Case : Open File Explorer

Use Case Name	Open File Explorer
Pre-condition	File Explorer must be available.
Basic Path	User says "Open File Explorer" → System opens File Explorer.
Alternate Path	User types "Open File Explorer".
Post-condition	File Explorer window opens.
Exception Path	If File Explorer is not available → Assistant shows an error.

3.3 Class Diagram

The UML Class Diagram represents the static structure of the AI-Based Virtual Assistant system. It showcases the system's major classes, their attributes and methods, the visibility scope of members, relationships among classes, and object-oriented principles like inheritance, aggregation, and generalization.[3]

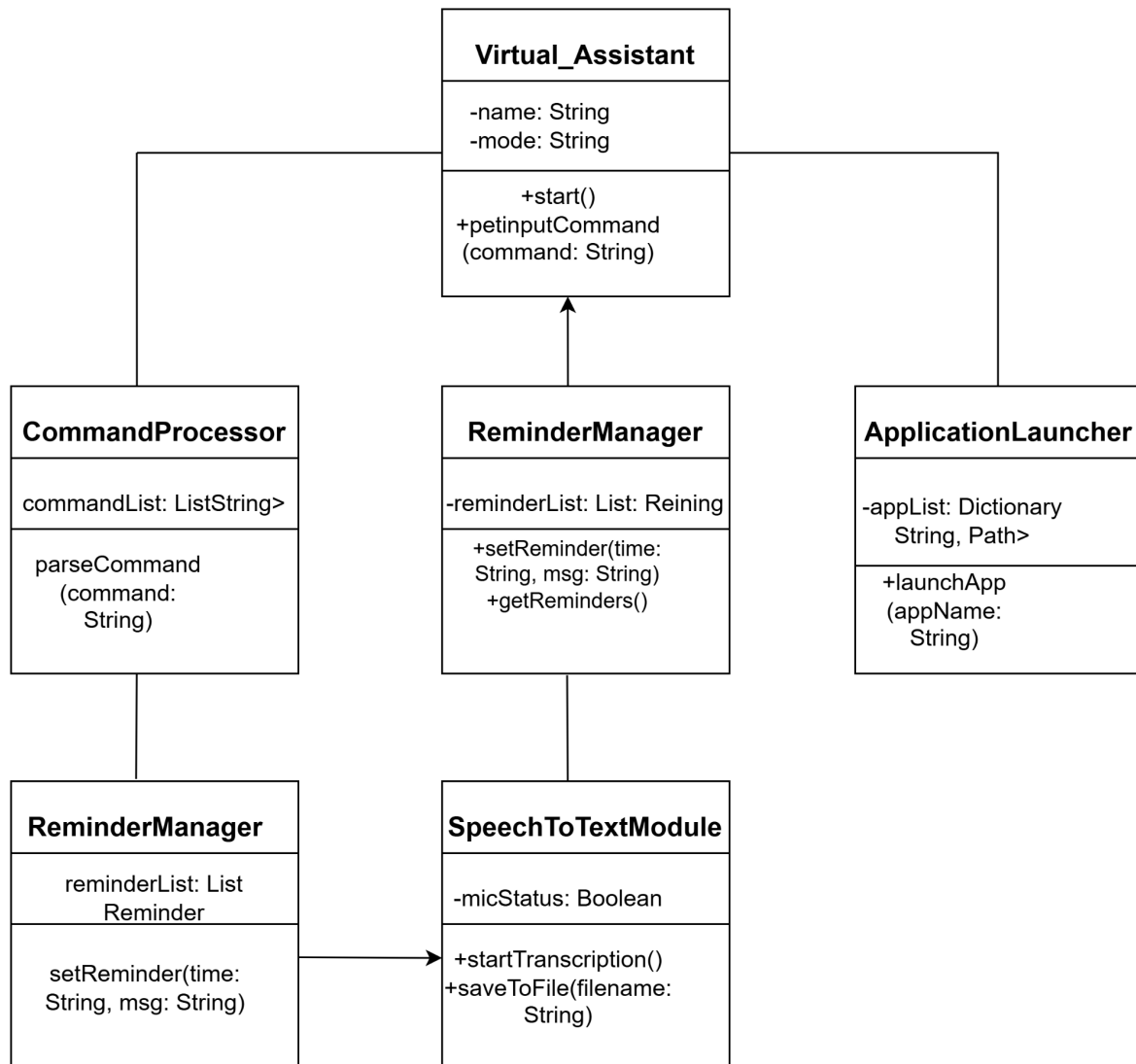


Figure 3.2 Class Diagram

3.3.1 Attributes

These are the data members of each class.

Example:

- user_input, output, date_time in VirtualAssistant
- voice_data in VoiceInput
- api_key in WeatherAPI

3.3.2 Methods

These are the functions that define class behavior.

Example:

- accept_input(), process_command() in VirtualAssistant
- listen() in VoiceInput
- get_weather_data() in WeatherAPI

3.3.3 Visibility Scope of Members

- **Private (-)**: Only accessible within the class (e.g., __user_input)
- **Protected (#)**: Accessible within the class and its subclasses (e.g., _assistant_name)
- **Public (+)**: Accessible from anywhere (e.g., start_game())

3.3.4 Associations

Represents a relationship between classes.

Example:

- VirtualAssistant is associated with VoiceInput, TextInput, Games (uses them but they can exist independently)

3.3.5 Generalization

It shows a general-parent class and a specific-child class relationship.

Example:

- VirtualAssistant generalizes (extends) AssistantBase

3.3.6 Inheritance

A class inherits attributes and methods from another class.

Example:

- VirtualAssistant inherits from AssistantBase, gaining common functionality

3.3.7 Aggregation

A “has-a” relationship where one class contains another, but both can exist independently.

Example:

- VirtualAssistant aggregates WeatherAPI and TaskManager
- TaskManager aggregates Task

3.4 Entity Relationship Diagram

An Entity - Relationship Diagram (ERD) is a visual representation of the entities within a system and the relationships between them. It is commonly used in database design to illustrate how data is structured and interrelated. ERD helps developers and analysts understand the data flow and how different components of the database interact.[2]

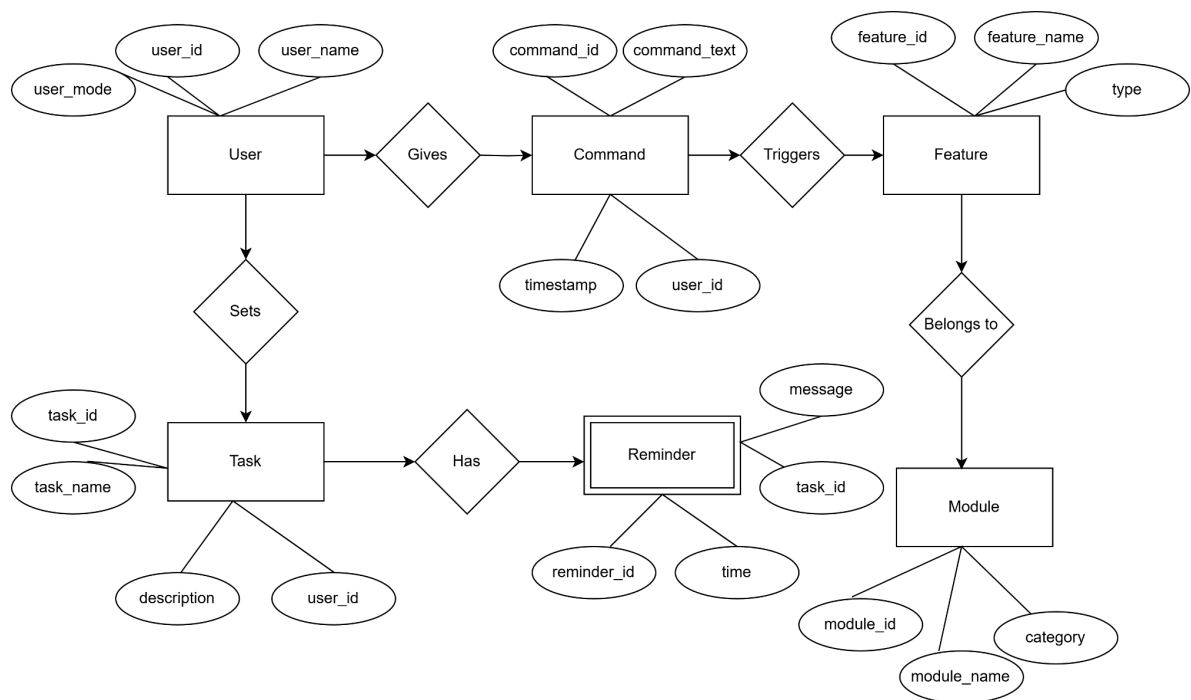


Figure 3.3 Entity Relationship Diagram

3.4.1 Entity(Tabular Format)

Entity Name	Type	Description
User	Strong	Represents the user interacting with the assistant
Command	Strong	The command given by the user
Feature	Strong	The functionality triggered by a command
Task	Strong	A user-created task (like note/reminder)
Reminder	Weak	A reminder linked to a task

3.4.2 Relationships(Tabular Format)

Entity 1	Relationship	Entity 2	Description
User	gives	Command	User gives a command
Command	triggers	Feature	Command triggers a feature
User	sets	Task	User sets a task (note, reminder, etc.)
Task	has	Reminder	A task may have one or more reminders
Feature	belongs to	Module	Feature belongs to a functional module (optional)

3.4.3 Attributes with Primary Key(Tabular Format)

Entity	Attributes	Primary Key
User	User_id, username, input_mode	user_id
Command	Command_id, command_text, timestamp	command_id
Feature	Feature_id, feature_name, type	feature_id
Task	Task_id, task, description, user_id (FK)	task_id
Reminder	Reminder_id, time, message, task_id(FK)	reminder_id

3.4.4 Cardinality(Tabular Format)

Relationship	Cardinality	Description
User — gives → Command	1 to Many	One user gives many commands
Command — triggers → Feature	Many to One	Many commands can trigger one feature
User — sets → Task	1 to Many	One user can set multiple tasks
Task — has → Reminder	1 to 1 or 1 to Many	A task may have one or multiple reminder
Feature — belongs to → Module	Many to One	Multiple feature can belong to one module

3.5 Activity Diagram

An Activity Diagram is used to model the dynamic aspects of the system, representing the flow of control from one activity to another. It helps visualize how the user interacts with the system and how tasks are carried out.[4]

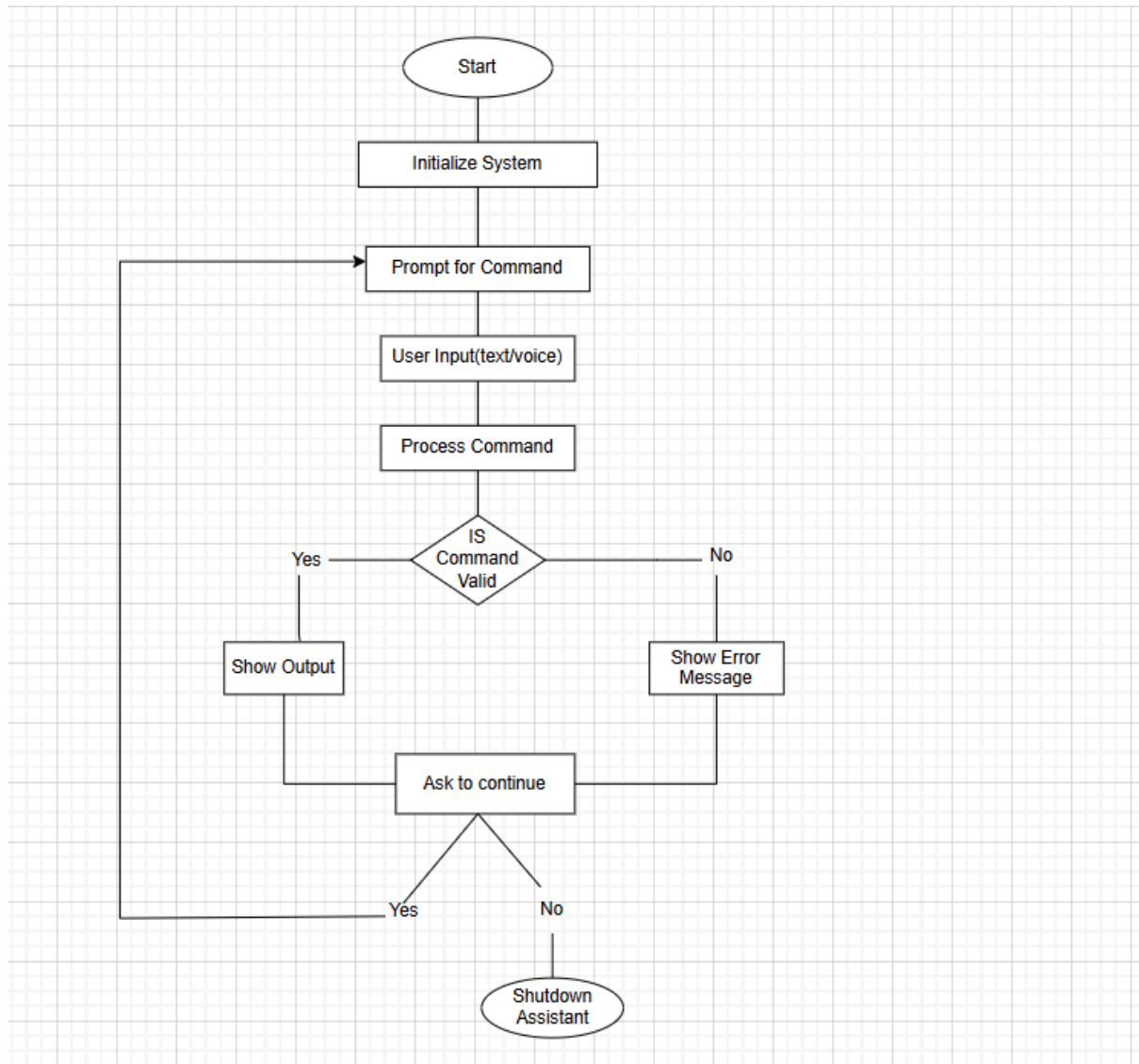


Figure 3.4 Activity Diagram

Chapter-4

Methodology

4.1 Descriptions of Different Input Validations and Checks

To ensure the reliability and robustness of the AI-Based Virtual Assistant, several input validations and checks have been incorporated. These checks help in preventing errors, handling exceptions gracefully, and improving the user experience by responding appropriately to different user inputs.

Below are the descriptions of the key input validations and checks implemented in the system:

4.1.1 Voice Input Availability Check

The system checks whether the microphone is active and able to capture voice input. If no speech is detected or the input is inaudible, the assistant prompts the user to repeat the command.

4.1.2 Empty Command Check

If the user provides no input or an empty string via voice or text, the assistant does not proceed with execution. Instead, it asks the user to provide a valid command.

4.1.3 Negative Command Recognition

The assistant is trained to recognize negative commands such as “don’t show date” or “do not open Chrome”. In such cases, the intended action is prevented, and the assistant confirms that the operation has been skipped as per the user’s instruction.

4.1.4 Unsupported Command Handling

When a user gives a command that is not supported or is outside the scope of the assistant’s functionalities, the system handles it gracefully by informing the user and suggesting supported actions.

4.1.5 Reminder Time Format Validation

While setting reminders, the system ensures that the time input is invalid numeric format (in seconds). Invalid formats or text-based time entries are rejected with a prompt for correct input.

4.1.6 Note Content Validation

When the user tries to create a note, the assistant checks if the input contains meaningful text. Blank or extremely short notes are considered invalid and are not saved.

4.1.7 Weather API and Joke API Error Handling

For web-based services such as weather reporting and joke fetching, the system includes exception handling to manage cases when APIs fail to respond or when there is no internet connectivity. A fallback response is provided to the user.[6],[7]

4.1.8 URL Format Validation for Web Operations

Before opening a website, search result, or game link, the assistant verifies whether the URL is valid and correctly formatted. Malformed URLs are ignored, and the user is informed.

4.1.9 System Compatibility Checks

Certain features, such as opening specific applications, depend on the operating system. The assistant checks the platform type (Windows, Linux, etc.) and uses system-specific commands to ensure compatibility.

4.1.10 File Path and Permission Check for Speech-to-Text Files

When saving transcribed speech as a text file, the system verifies whether the destination path is accessible and whether the application has write permissions. This prevents accidental overwriting or permission-related errors.

4.2 Sample Test Cases

Sample test cases are designed to evaluate the functional correctness of different modules of the system. These test cases help ensure that the virtual assistant performs accurately according to the specified input commands and produces the expected output. Two types of software testing techniques have been applied: **White Box Testing** and **Black Box Testing**.

4.3 Sample White Box Test Cases

White Box Test Cases are test cases designed based on the internal structure, logic, and code of the application being tested. Unlike black-box testing, where the focus is on the functionality of the system from the user's perspective, white-box testing examines the internal workings, such as functions, control flow, and decision paths.

4.3.1 Flow Graph

A flow graph represents the logic flow of a particular module or function. It is used to visually understand the program's structure. For white-box testing, you need to create flow graphs for each module.[1]

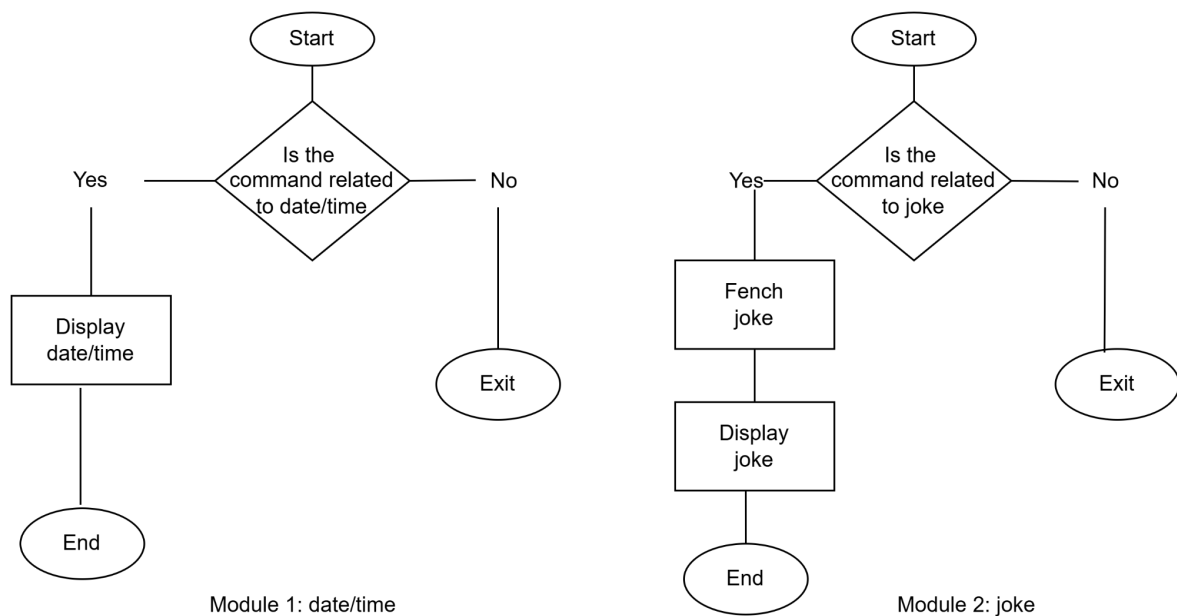


Figure 4.3.1 Flow Graph

4.3.2 Cyclomatic Complexity Calculation (Module 1: Date/Time)

Total Nodes (N) = 5

Total Edges (E) = 5

Number of connected components (P) = 1

Formula:

$$V(G) = E - N + 2P$$

Substitute:

$$V(G) = 5 - 5 + 2(1) = 2$$

Cyclomatic Complexity = 2

4.3.3 Independent Paths

Path_id	Description
P1	1 → 2 → 3 → 4 (User asked for date/time)
P2	1 → 2 → 5 → 6 (User command not related to date)

4.3.4 Test Case Execution

Test Case_id	Path	Input Command	Expected output	Actual Output	Result
TC1	P1	What is time	System display time	Time shown	Pass
TC2	P2	What is date	System display date	Date Shown	Pass

4.3.5 Cyclomatic Complexity Calculation (Module 2: Joke)

Total Nodes (N) = 6

Total Edges (E) = 6

Formula:

$$V(G) = E - N + 2P$$

Substitute:

$$V(G) = 6 - 6 + 2(1) = 2$$

Cyclomatic Complexity = 2

4.3.6 Independent Paths

Path_id	Path Description
P1	1 → 2 → 3 → 4 → 6 (User asked for a joke)
P2	1 → 2 → 5 → 6 (User command not related to joke)

4.3.7 Test Case Execution

Test Case_id	Path	Input Command	Expected Output	Actual Output	Result
TC1	P1	Tell me a joke	A joke is fetched and shown	Joke displayed	Pass
TC2	P2	Open Chrome	Input not matched with joke module; no action taken	Ignored by this module	Pass

4.4 Sample Black Box Test Cases

Black box testing focuses on verifying the functionality of the system without looking into its internal code structure. Here, the system is tested by giving various inputs and validating outputs based on requirements.

4.4.1 Date and Time Module (Equivalence Class Testing)

Test Case_id	Input Command	Class Type	Expected Output	Result
TC1	What's the date	Valid	Displays current date	Pass
TC2	What time is it	Valid	Display current time	Pass
TC3	Open Chrome	Invalid	No response	Pass

4.4.2 Joke Module(Equivalence Class Testing)

Test Case_id	Input Command	Class Type	Expected Output	Result
TC1	Tell me a joke	Valid	Joke displayed	Pass
TC2	Open notepad	Invalid	No response	Pass

Chapter-5

Implementation

5.1 Description of Different Reports

This section outlines the different types of reports generated during the development and operation of the AI-Based Virtual Assistant. These reports help in analyzing performance, diagnosing issues, and improving the system.

5.1.1 Command Execution Report

This report maintains a log of all the commands executed by the assistant. It helps in verifying whether a command was recognized correctly and performed as expected.

5.1.2 Error Handling Report

This report documents any errors or exceptions that occur during the execution of tasks. It is useful for debugging and improving system reliability.

5.1.3 System Performance Report

This report provides insights into the system's performance, including CPU and memory usage while the assistant is running. It ensures the system operates efficiently without overloading resources.

5.1.4 Voice Recognition Accuracy Report

This report evaluates the accuracy of the speech recognition module. It identifies how accurately the assistant understands user commands and highlights areas for improvement.

5.1.5 API Response Report

This report logs the responses received from external APIs used by the assistant, such as for weather, news, or jokes. It helps in monitoring the correctness and timeliness of data.

5.1.6 Task Execution Time Report

This report records the time taken to complete various tasks initiated by the user. It is used to measure the responsiveness of the assistant.

5.1.7 User Feedback Report

This report collects user feedback based on their experience with the assistant. It is useful for analyzing satisfaction levels and identifying feature requests or issues.

5.1.8 Feature Usage Report

This report tracks how frequently different features of the assistant are used. It helps in understanding user preferences and guiding future improvements.

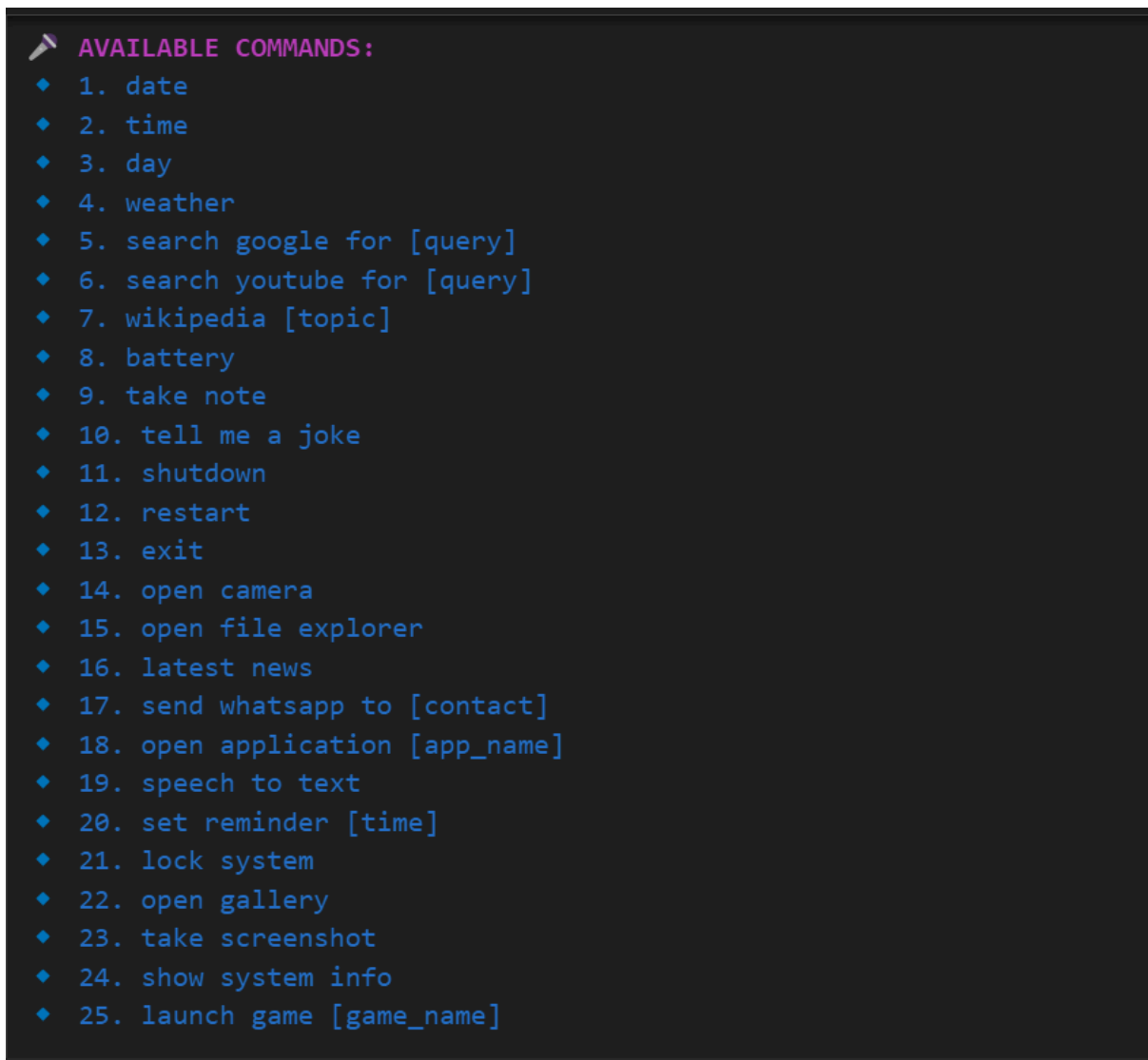
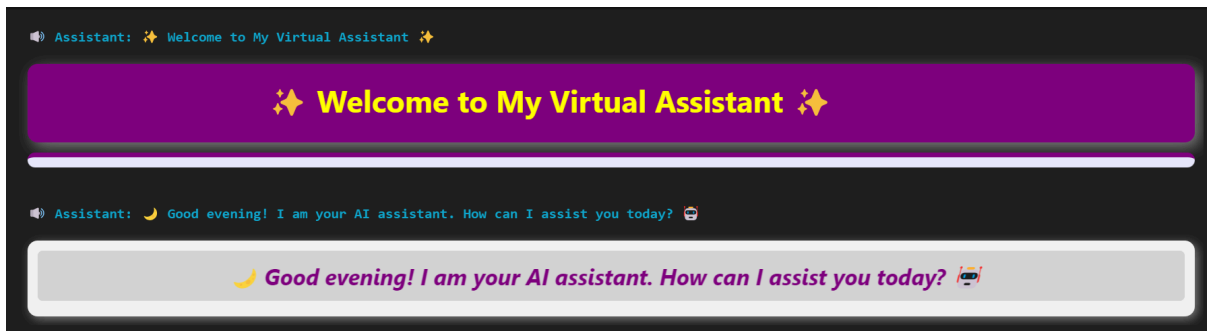
5.1.9 Scheduled Task Report

This report keeps a record of all the tasks that have been scheduled to run at a specific time. It confirms whether the scheduled actions were executed successfully.

5.1.10 System Logs Report

This comprehensive report includes all major activities performed by the assistant. It acts as a centralized log for monitoring system behavior and auditing usage.

5.2 Snapshots of the different input and output screens



🔊 Assistant: Would you like to type or speak your command? Please say or type 'text' or 'voice'.
🔊 Assistant: Please speak...
🎧 **Listening...**
🗣️ You said: please show me today date
🔊 Assistant: Processing your request: please show me today date
🔊 Assistant: Today's date is 2025-04-21.

🔊 Assistant: Please speak...
🎧 **Listening...**
🗣️ You said: please show me current time
🔊 Assistant: Processing your request: please show me current time
🔊 Assistant: Current time is 12:40 AM.

🔊 Assistant: Please speak...
🎧 **Listening...**
🗣️ You said: day today
🔊 Assistant: Processing your request: day today
🔊 Assistant: Monday

🎧 **Listening...**
🗣️ You said: show me weather detail
🔊 Assistant: Processing your request: show me weather detail
Delhi weather:
Condition: Clear Sky
Temperature: 34.42°C
Humidity: 14%
Wind Speed: 0.66 m/s

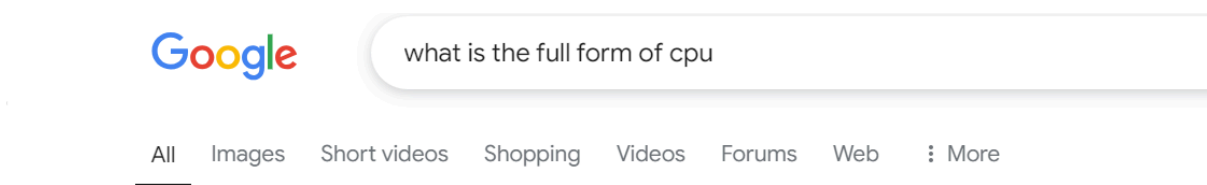
🔊 Assistant: Please speak...
 🎧 Listening...
 🗣️ You said: Google for

🔊 Assistant: Processing your request: google for

🔊 Assistant: What would you like to search on Google? You can say or type.

🔊 Assistant: Please speak...
 🎧 Listening...
 🗣️ You said: what is the full form of CPU

🔊 Assistant: Searching Google for what is the full form of cpu



CPU › Full name :

Central processing unit



🔊 Assistant: Would you like to type or speak your command? Please say or type 'text' or 'voice'.
 🎧 Listening...
 🗣️ You said: Wikipedia topic about Java

🔊 Assistant: Processing your request: wikipedia topic about java

🔊 Assistant: What would you like to search on Wikipedia? You can say or type.

🔊 Assistant: Please speak...
 🎧 Listening...
 🗣️ You said: when java comes who developed Java

🔊 Assistant: According to Wikipedia: Java is a set of computer software and specifications that provides a software platform for developing application software and deploying it in a cross-platform computing environment. Java is used in a wide variety of computing platforms from embedded devices and mobile phones to enterprise servers and supercomputers.

🔊 Assistant: Please speak...
🎧 Listening...
🗣️ You said: battery percentage

🔊 Assistant: Processing your request: battery percentage

🔊 Assistant: Battery is at 67 percent.

🔊 Assistant: Please speak...
🎧 Listening...
🗣️ You said: tell me a joke

🔊 Assistant: Processing your request: tell me a joke

🔊 Assistant: Did you hear about the Mexican train killer? - He had loco motives
😄 Joke: Did you hear about the Mexican train killer? - He had loco motives

🔊 Assistant: Please speak...
🎧 Listening...
🗣️ You said: please lock my system

🔊 Assistant: Processing your request: please lock my system

🔊 Assistant: Locking the system.

🔊 Assistant: Please speak...
🎧 Listening...
🗣️ You said: please show me my system info

🔊 Assistant: Processing your request: please show me my system info

🔊 Assistant: Here is the system information:

```
System: Windows
Node: DESKTOP-9A2J8V7
Release: 10
Version: 10.0.26100
Machine: AMD64
Processor: Intel64 Family 6 Model 142 Stepping 12, GenuineIntel
CPU Usage: 11.0%
RAM: 61.1% used
```

Assistant: Please speak...

Listening...

You said: please search YouTube for query

Assistant: Processing your request: please search youtube for query

Assistant: What would you like to search on YouTube? You can say or type.

Assistant: Please speak...

Listening...

You said: please give me playlist of DSA

Assistant: Searching YouTube for please give me playlist of dsa

Chapter 1
DSA Series
by Shradha Ma'am

- Flowcharts & Pseudocode
- Installation

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Lecture 2 : Variable, Data Types & Operators | DSA Series by Shradha Ma'am | C++ · 1:16:44

[View full playlist](#)

Assistant: Please type your command.

You typed: speech to text

Assistant: Processing your request: speech to text

Assistant: Please speak. I will convert your speech into text.

Listening for speech-to-text input...

Assistant: Your speech has been saved to speech_to_text_20250421_012815.txt

Transcription saved: speech_to_text_20250421_012815.txt

Text: everyone my name is Yash

You said: set reminder for prayer

Assistant: Processing your request: set reminder for prayer

Assistant: What should I remind you about?

Assistant: In how many minutes should I remind you?

Assistant: Okay, I will remind you in 2 minute(s).

Reminder set for 2 minute(s): Prayer time

🔊 Assistant: Please speak...

🔊 Listening...

🗣️ You said: please show me latest news

🔊 Assistant: Processing your request: please show me latest news

🔊 Assistant: Opening the latest news for you...

🔊 Assistant: Here are the top trending news headlines.

Top stories >



The Hindu

Nishikant Dubey slams former CEC Quraishi, calls him 'Muslim commissioner'

3 hours ago

Times of India

'Muslim commissioner': Amid Supreme Court row, BJP's Nishikant Dubey targets poll panel ex-chief

5 hours ago

The Hindu

Letters to The Editor — April 21, 2025

58 minutes ago • Opinion

The Indian Express

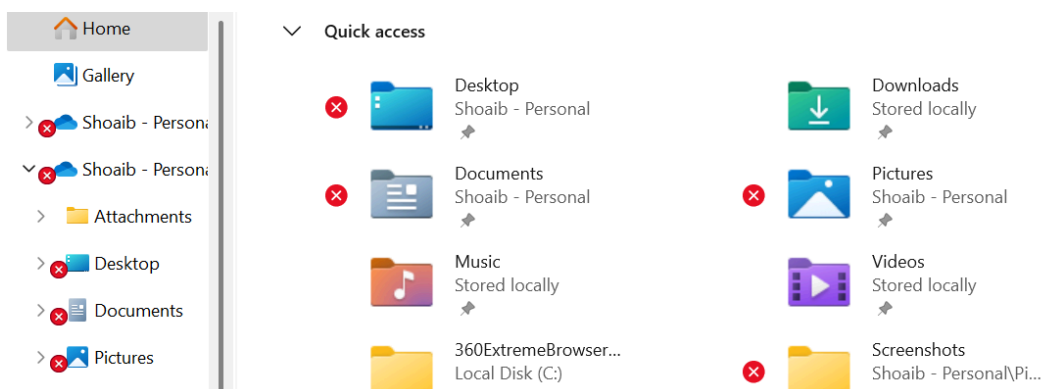
The other BJP leader who left party red-faced with remarks on judiciary: Who is Dinesh Sharma?

7 hours ago • By Lalmani Verma

Full coverage

🗣️ You said: open file explorer

🔊 Assistant: Processing your request: open file explorer



🔊 Assistant: Please speak...
🎧 **Listening...**
🗣️ You said: launch game snake

🔊 Assistant: Processing your request: launch game snake

🔊 Assistant: Launching online Snake game.



🔊 Assistant: Please type your command.
📄 **You typed: whatsapp 8434914916 Hello**

🔊 Assistant: Processing your request: whatsapp 8434914916 hello

🔊 Assistant: Message sent to 8434914916

🔊 Assistant: Please speak...
🎧 **Listening...**
🗣️ You said: you can please restart my system

🔊 Assistant: Processing your request: you can please restart my system

🔊 Assistant: Restarting system.

🔊 Assistant: Please speak...

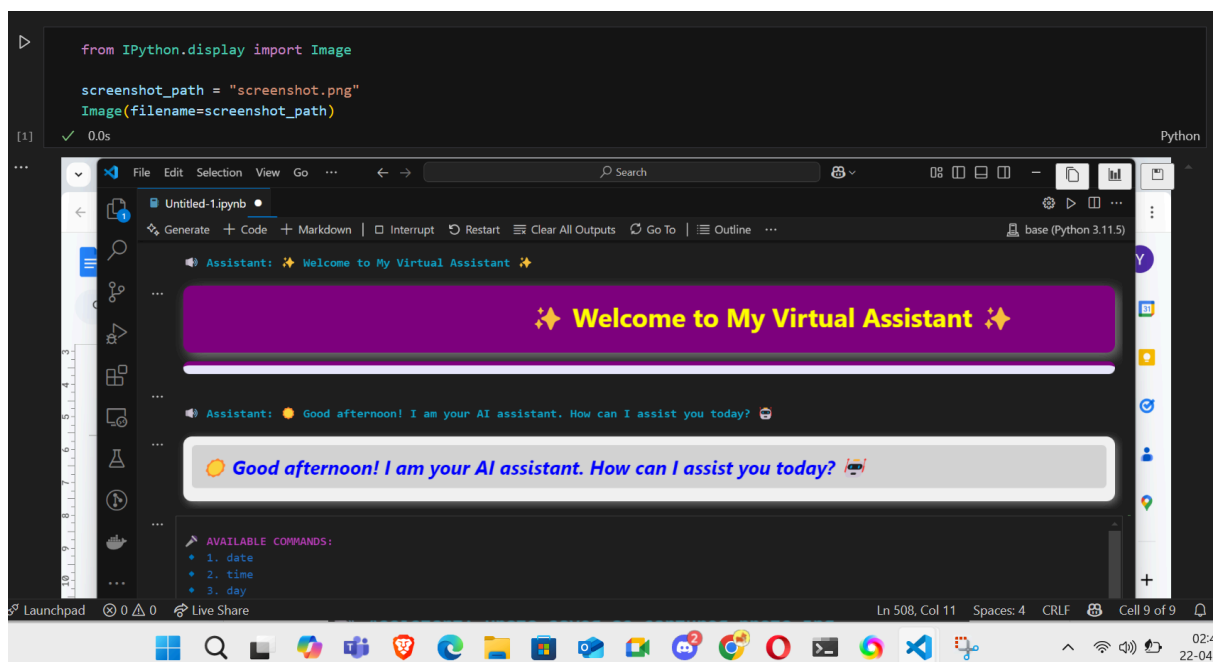
🔊 Listening...

🔊 You said: please take screenshot

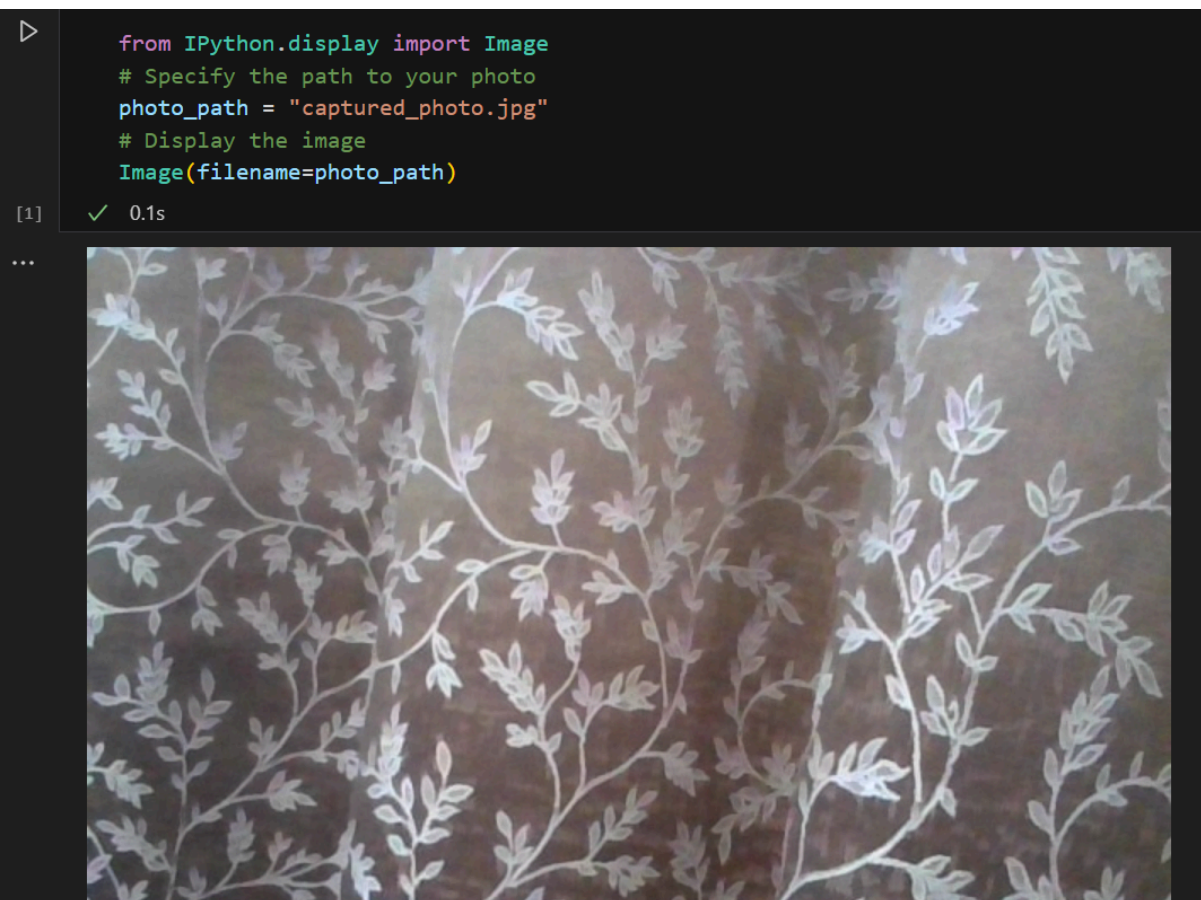
🔊 Assistant: Processing your request: please take screenshot

🔊 Assistant: Screenshot has been taken and saved.

📷 Screenshot saved as screenshot.png



Assistant: Please speak...
Listening...
You said: please open camera
Assistant: Processing your request: please open camera
Assistant: Photo saved as captured_photo.jpg
Photo saved as captured_photo.jpg



Assistant: Please speak...
Listening...
You said: exit the program
Assistant: Processing your request: exit the program
Assistant: Goodbye! Have a nice day!

Chapter-6

Result and Discussion

The AI-Based Virtual Assistant utilizes various features to assist users, such as voice/text command recognition, system control, and web interaction. The system's performance is assessed based on how accurately it responds to these commands. This chapter discusses the advantages and challenges faced during the implementation and the practical results observed.[9]

6.1 Advantages

- **Enhanced User Experience**

The virtual assistant offers seamless interaction through both voice and text, improving user convenience and accessibility.

- **Multifunctional Capabilities**

It performs a wide range of tasks like managing reminders, telling jokes, providing weather updates, opening apps, fetching news, and more, making it a comprehensive tool.

- **Easy System and Web Integration**

The assistant can open various applications, conduct web searches, and execute system commands, making it a versatile solution for users.

- **Voice/Text Command Recognition**

The system is able to understand natural language through voice or text input, making it more user-friendly and intuitive.

- **API Integration**

The assistant integrates external APIs for functionalities like weather updates and joke fetching, providing dynamic, real-time information.

- **Customizable and Extensible**

The project is flexible, and new features or commands can easily be added in the future, allowing it to grow and adapt to new user needs.

- **Cross-Platform Compatibility**

The system checks the platform type (Windows, Linux) to ensure that system-specific commands are used for compatibility.

- **Error Handling and Input Validation**

The assistant has robust error handling and input validation checks that ensure smooth operation, preventing miscommunication or system failures.

6.2 Disadvantages

- **Limited OS Support**

The assistant may not work seamlessly on all operating systems due to dependencies on platform-specific commands. It primarily supports Windows and Linux environments.

- **Dependency on Internet Connectivity**

Features like weather updates, joke fetching, and certain web searches rely on an active internet connection, limiting functionality in offline modes.

- **Speech Recognition Accuracy**

While the assistant performs well in ideal conditions, background noise or unclear speech may reduce the accuracy of voice recognition.

- **Limited NLP Capabilities**

The assistant's natural language processing (NLP) is limited to simple tasks and may struggle with more complex or nuanced user queries.

- **Limited Knowledge Base**

The assistant's knowledge base may be limited to predefined APIs and functionalities, meaning it cannot answer complex or off-topic questions outside its programming.

- **No Mobile Application**

Currently, the assistant is a desktop-based solution, and there is no mobile app version, limiting access to users who prefer mobile devices.

- **Potential API Downtime**

Since the assistant relies on third-party APIs (e.g., weather and jokes), any downtime or failure from the API providers can result in missing functionality.

- **User Command Restrictions**

The assistant's ability to understand complex commands or ambiguous language may be limited, requiring users to issue very specific instructions for accurate results.

Chapter-7

Conclusion and Future Scope

7.1 Conclusion

The AI-Based Virtual Assistant project has successfully implemented a multifunctional tool designed to streamline system operations and enhance user interaction. By integrating voice and text command recognition, the assistant is capable of performing a variety of tasks, including telling the date and time, managing reminders, opening applications, fetching news, and providing weather updates, all while ensuring a smooth user experience.

The project showcases a robust error-handling mechanism and input validation to ensure reliability and consistency in performance, even when faced with unexpected inputs or failures in external APIs. Additionally, the assistant's modular design allows for future enhancements, including the potential for mobile compatibility, more intelligent natural language processing (NLP), and broader integration with external services and smart home devices.

Despite the system's many strengths, there are certain limitations such as its dependency on internet connectivity, potential issues with speech recognition in noisy environments, and platform-specific compatibility. These limitations provide valuable insights for future improvements.

Overall, this project demonstrates the potential of AI and voice-based assistants in improving everyday tasks and system operations. As technology continues to evolve, the assistant's capabilities can be expanded further, making it a more powerful and essential tool for users. The project's success serves as a strong foundation for further research and development in the field of virtual assistants.[10]

7.2 Future Scope

The future scope of the AI-Based Virtual Assistant project presents many exciting opportunities for expansion, making the system more intelligent, adaptable, and integrated with modern technologies. These enhancements will allow the assistant to serve a wider range of users and adapt to their evolving needs.[11]

7.2.1 Mobile Application Development

The assistant can be expanded into a mobile application, enabling users to interact with it from smartphones and tablets, enhancing accessibility.

7.2.2 Multi-Language Support

The assistant can be enhanced to understand and respond in multiple languages, broadening its user base and making it more adaptable for global use.

7.2.3 Advanced Natural Language Processing (NLP)

Integrating more advanced NLP techniques will enable the assistant to understand and process more complex commands and carry out more nuanced conversations, enhancing its usability.

7.2.4 Smart Home Integration

By integrating the assistant with IoT (Internet of Things) devices, it could control smart home systems like lighting, heating, security systems, and more, offering users greater convenience.

7.2.5 Cloud Storage and Synchronization

Cloud-based storage can be added for saving notes, reminders, and transcribed text files. This will allow users to access their data from multiple devices, ensuring seamless synchronization.

7.2.6 Personalized User Experience

The assistant can be enhanced to learn from user interactions and offer more personalized responses, remembering preferences, frequently used applications, and commands.

7.2.7 Voice Recognition Improvements

Improving the accuracy and robustness of voice recognition, especially in noisy environments, can help the assistant function better in diverse real-world scenarios.

7.2.8 Integration with More Web Services

Future versions could integrate additional services like social media management, stock market tracking, and more, expanding the assistant's functionality to include a wide range of tasks.

7.2.9 Security Features

The project could include advanced security features, such as biometric authentication (e.g., face or fingerprint recognition) to protect sensitive data and actions within the assistant.

7.2.10 Integration with AI and Machine Learning

Implementing machine learning algorithms could allow the assistant to predict user needs, improving its efficiency over time and creating a more intelligent user experience.

7.2.11 Enhanced Error Handling

Future development could focus on improving error recovery and providing more helpful guidance to users when errors or unsupported commands occur.

7.2.12 Support for Multiple Platforms

Expanding the assistant to support additional platforms (e.g., macOS, Android, or other operating systems) would increase its reach and usability.

Chapter-8

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